

Mine One

Multiplayer using Unity Netcode for GameObjects



RIYA

2310991192

HOW DOES MY GAME STANDOUT ?

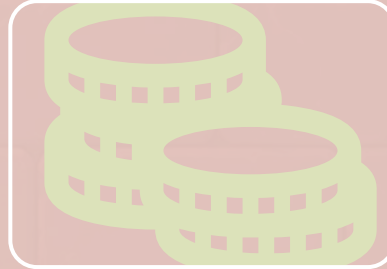
Unlike traditional Minesweeper, my game is a **real-time multiplayer competition**.



Everyone interacts with the **same board**.



Players race to reveal numbers.



Every number revealed gives points.



The faster you play, the more points you can earn.



Reveal a **bomb** → **you die.**

Core Goal: Score as many points as possible without hitting a bomb.



Main Functionalities

Multiplayer Lobby System

Multiplayer lobby
using Unity
Netcode for
GameObjects.

Host can create a
lobby.

Other player can
join using a **join
code**.

Displays lobby
join code to
players.

Prevents starting
until at least 2
players are
present.

Player Name System

Players enter
their name before
hosting/joining.

Name is stored
using
PlayerPrefs.

Name persists
between
sessions.

Validation
prevents empty
player names.

riya

Name is compulsory to set before game creation or joining.

◆ CREATE GAME ◆

◆ OR ◆

Enter code

◆ JOIN GAME ◆

Host creates
the game.

Host teleported
to lobby.

riya

0

JOIN CODE

KCJ7GK

Waiting for
players...

MAP SIZE



5



Playerrrrr

Joining lobby...

◆ CREATE GAME ◆

OR

kcj7gk

◆ JOIN GAME ◆

Client
enters Join
Code

Client
teleported to
lobby.

riya

0

Playerrrrr

0

JOIN CODE

KCJ7GK

◆ Start Game ◆

MAP SIZE



5



Dynamic Minesweeper Grid

The game is **round-based**, rather than having a fixed number of levels.

The host can choose a new **map size after every round**, keeping each round unpredictable.

Grid sizes range from **5×5 to 100×100**, with automatic cell resizing.

There is **no fixed level limit** — the challenge is about **how far players can keep going**.

Increasing or changing the map size allows the difficulty to adapt naturally as players progress.

This makes the game **more replayable and competitive** than a fixed map or limited progression system.

Each round creates a new challenge, so players cannot simply memorize a fixed level layout or progression path.

Server-Authoritative Game

The server controls the actual game state.

Mine generation happens on the server.

Player clicks are sent to the server through RPCs.

Prevents individual clients from independently modifying the game state.



Each player has:

- Name
- Color
- Score

Player information is
synchronized across the
network.

riya	0	★
Playerrrrr	0	★

Unrevealed Map

Random Mine Generation

Mines are randomly generated when the **first cell is clicked**.



The first clicked cell and its surrounding 8 cells are guaranteed to be safe.



Approximately **20% of eligible cells** become bombs.

Minesweeper Number Calculation

Each safe cell calculates how many bombs are in its surrounding 8 cells.



Numbers are displayed on revealed cells.



Number colors correspond to the player who revealed the cell.

Multiplayer Cell Claiming

Players compete to reveal cells.



Each revealed cell records **which player revealed it.**



Players can visually see ownership through their assigned color.

Automatic Empty-Cell Expansion

Uses a stack-based flood-fill system.



When a player reveals a cell with **0 adjacent bombs**, surrounding safe cells are automatically revealed.



Continues recursively until numbered cells are reached.

Scoring System

Players earn points based on the number of bombs surrounding the cells they reveal.



Player scores are synchronized across all clients.



Scores are displayed beside each player's name.

Bomb / Disqualification System

Clicking a bomb causes the player to become **Disqualified**.

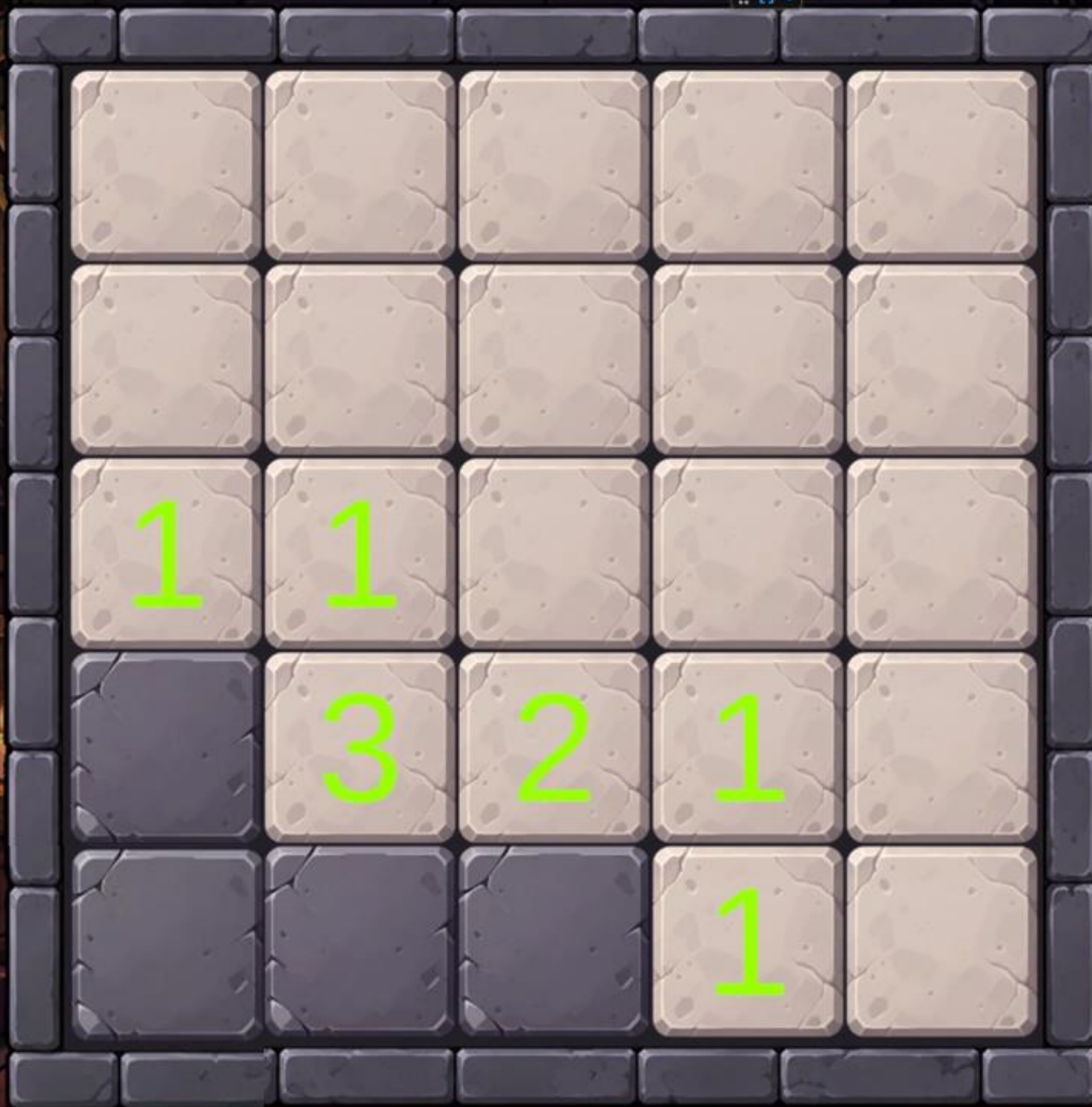
Their player card becomes visually disabled/greyed out.

A warning message announces which player was disqualified.

If All safe cells are uncovered or all players are disqualified, the game ends.

The game tracks the number of remaining playable players.

Disqualified players can no longer interact with the board.



The numbers you reveal adds up to your score





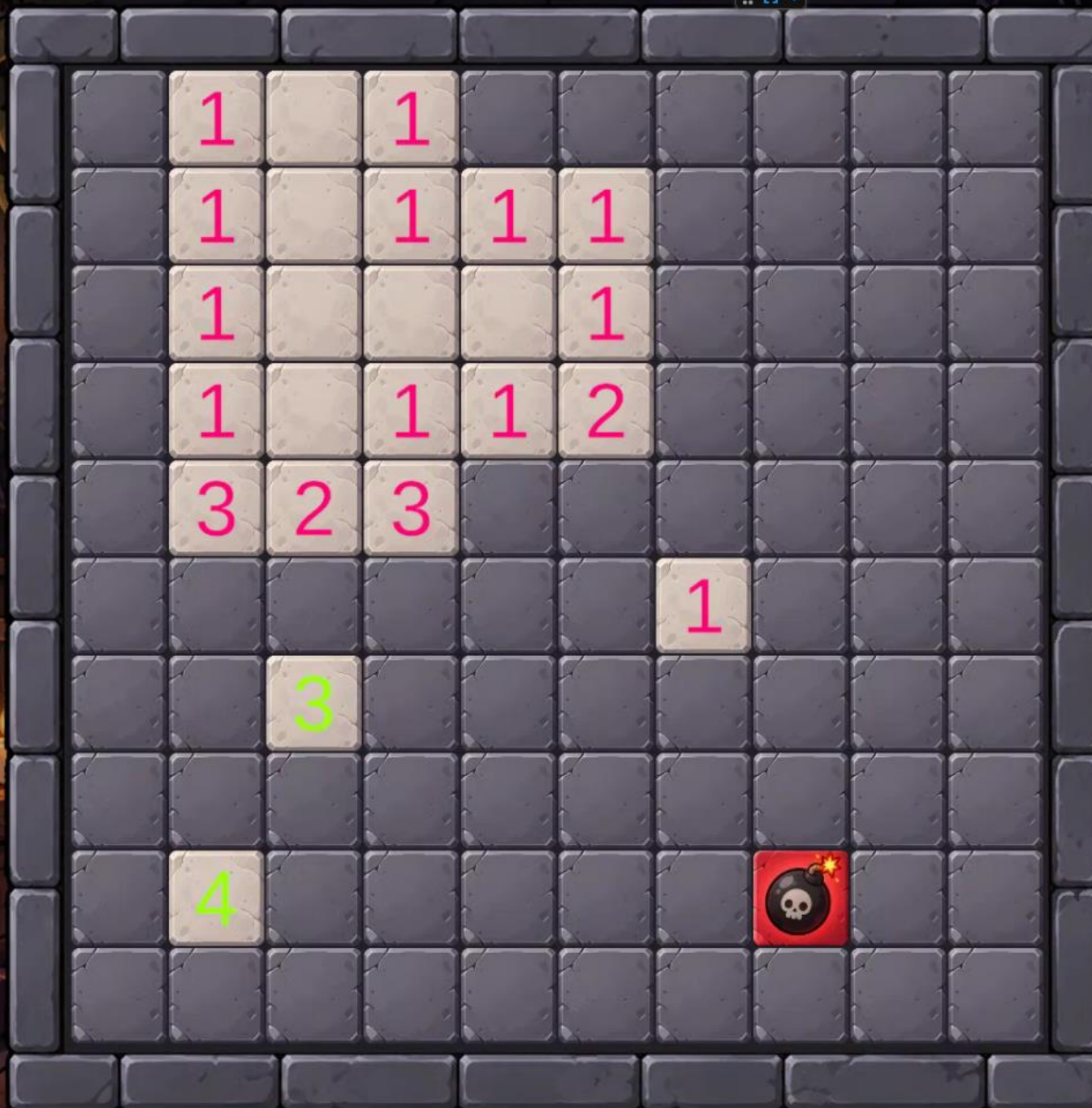
riya	9
Playerrrrr	0

If all cells are revealed
and no bomb is found,
all players teleport
back to lobby for next
round.

KCJ7GK

◆ Start Game ◆

 **5** 



riya has been disqualified!

Player disqualified after bomb is found.

riya	16	★
Playerrrrr	22	★

The players left can still continue the round to see how far they can achieve.

When all players are disqualified, they are teleported to the result screen.

1	Playerrrrr	42
2	riya	16

◆ Play Again ◆

Play again sends players
to lobby where you can
start fresh.

riya

0

Playerrrrr

0

KCJ7GK

◆ Start Game ◆



5



Network Synchronization

Uses **Netcode RPCs** and **NetworkVariables/NetworkLists** to synchronize:

Player
names

Scores

Player
states

Player
colors

Grid size

Join code

Cell
states

Revealed
cells

Bomb
results



THANKYOU